



程序设计原理

程序设计概述和c++基础知识

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课程目标

掌握结构化程序设计方法；选择合适的开发环境、工具，并能熟练使用该工具完成程序设计工作；培养学生的逻辑思维能力，培养其严谨的思维方式和良好的程序设计风格。

- 不仅是一门程序设计语言课程
- 思维课程：培养逻辑思维能力
- 以C++程序设计语言为例，培养逻辑思维能力



学习方法

实践、实践、再实践，实践是检验真理的唯一标准

纸上学来终觉浅，绝知此事要躬行
纵使思忖千百度，不如亲手下地锄
雄关漫道真如铁，而今迈步从头越

有天赋的不要骄傲，可以**适度**帮助同学
没有天赋的**不要怕、不要怨、不要愁**，踏踏实实，
勤勤恳恳，任劳任怨、归根到底一个字就是：**练**！



C++ Development Tools

➤ For Windows

- l CodeBlocks

- l Dev-C++

- l VS Code

➤ For Linux

- l CodeBlocks

Homework: Download the software and install it.

Notes: There are serval different versions of these software on the site. You may choose which version is best for you.



Framework of C++ Program

```
#include <iostream>
```

```
using namespace std;
```

```
/*This is my first program! It's too simple , but I hate you .You make me so busy  
and crazy !*/
```

```
int main()
```

```
{
```

```
    cout << "Welcome to C++!" << endl; //comments bala bala
```

```
    cout << "Welcome to Zhisuan!" << endl;
```

```
    cout << "Welcome to Tianjin University! " << endl;
```

```
    return 0;
```

```
}
```



Exercise

Write a program which prints "Hello World" to standard output.

Input

There is no input for this problem.

Output

Print "Hello World" in a line.

Sample Input 1

No input

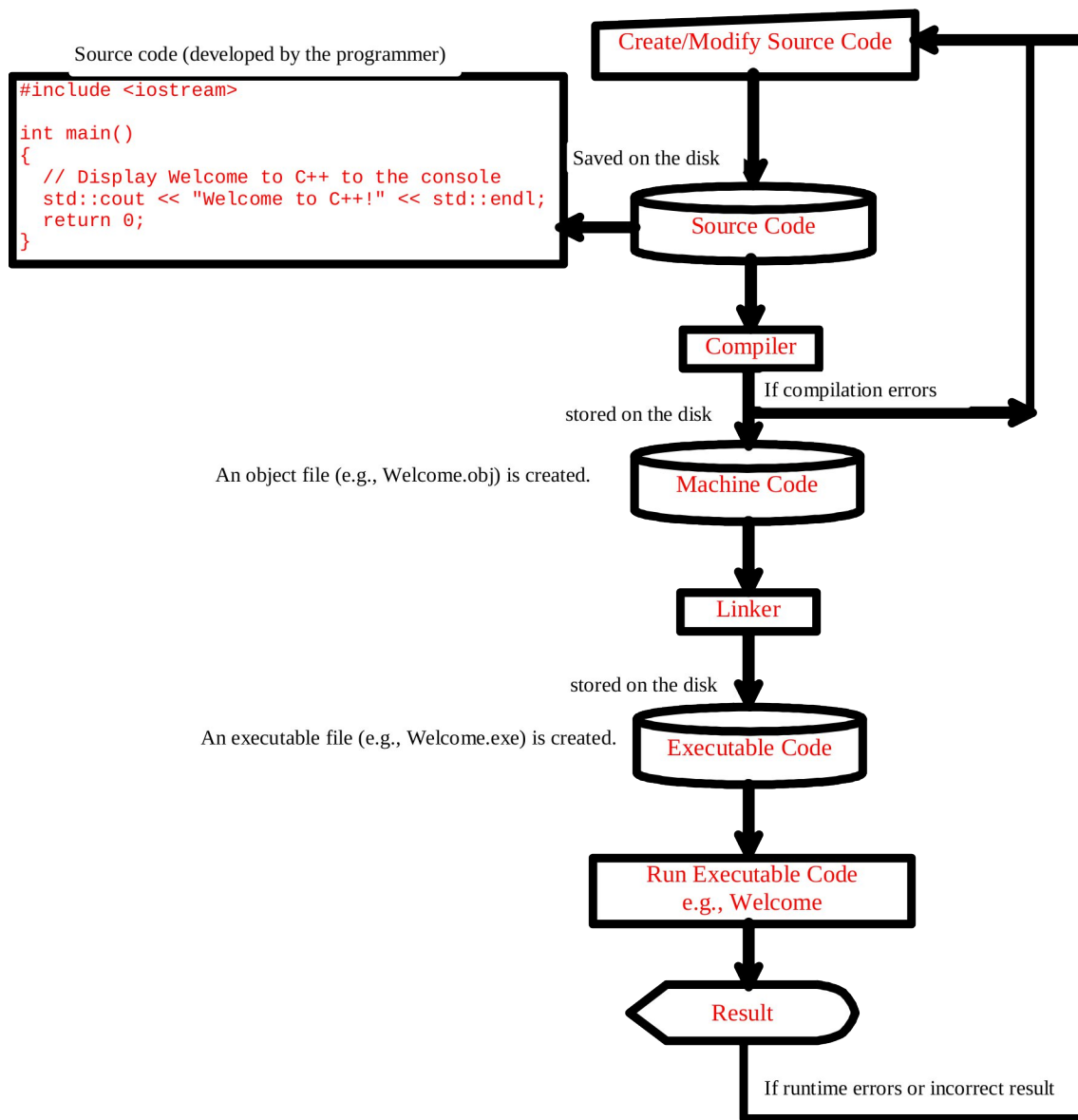
Sample Output 1

Hello World



Program running process

Creating, Compiling, and Running Programs





Programming Style and Documentation

- **Appropriate Comments and Comment Styles**
- **Proper Indentation and Spacing**

```
#include <iostream>  
using namespace std;
```

```
/*This program preform mathematical computation and display  
the result on the console.*/
```

```
int main()  
{  
    cout << "(10.5 + 2 * 3) / (45 - 3.5) = ";  
    cout << (10.5 + 2 * 3) / (45 - 3.5) << endl;  
  
    return 0;  
}
```




Identifiers

- An identifier is a sequence of characters that consists of letters, digits, and underscores (_).
- An identifier must start with a letter or an underscore. It cannot start with a digit.
- An identifier cannot be a reserved word.
- An identifier can be of any length, but your C++ compiler may impose some restriction. Use identifiers of 31 characters or fewer to ensure portability.



Declaring Variables

```
int x,y;           // Declare x,y to be  
                  // integer variables;  
double radius;    // Declare radius to  
                  // be a double variable;  
char a;           // Declare a to be a  
                  // character variable;
```



Assignment Statements

```
x = 1;           // Assign 1 to x;
y = 2;           // Assign 2 to y;
radius = 1.0;    // Assign 1.0 to radius;
a = 'A';         // Assign 'A' to a;
x = y + 1;
//Assign the addition of y and 1 to x;
x = x + 1;
int a = 1;
double d = 1.4;
```



Standard output and input

✓ cout

```
#include <iostream>
using namespace std;
int main()
{
    int a = 10;
    double r = 2.0;
    int s = 32767;

    cout << s << endl;
    cout << r * r << endl;

    return 0;
}
```

ü cin

```
#include <iostream>
using namespace std;
int main()
{
    double r = 2.0;
    int s = 32767;

    cin >> s >> r;

    cout << s << endl;
    cout << r * r << endl;

    return 0;
}
```



Mathematical Functions

➤ Trigonometric Functions

cos	Compute cosine (function)
sin	Compute sine (function)
tan	Compute tangent (function)
acos	Compute arc cosine (function)
asin	Compute arc sine (function)
atan	Compute arc tangent (function)
atan2	Compute arc tangent with two parameters (function)

$\sin(0)$ returns 0.0

$\sin(270 * \text{PI}/180)$ returns -1.0

$\sin(\text{PI} / 6)$ returns 0.5

$\cos(\text{PI} / 6)$ returns 0.866

$\text{asin}(0.5)$ returns 0.523599($\text{PI}/6$)



Mathematical Functions

➤ Exponent Functions

exp	Compute exponential function (function)
log	Compute natural logarithm (function)
log10	Compute common logarithm (function)
pow	Raise to power (function)
sqrt	Compute square root (function)

`exp(1.0)` returns 2.71828

`log(E)` returns 1.0

`log10(10.0)` returns 1.0

`pow(2.0,3)` returns 8.0

`sqrt(4.0)` returns 2.0



Mathematical Functions

➤ Rounding Function

ceil	Round up value (function)
floor	Round down value (function)
fmod	Compute remainder of division (function)

double ceil (double x);

ceil of 2.3 is 3.0

ceil of 3.8 is 4.0

ceil of -2.3 is -2.0

ceil of -3.8 is -3.0

double floor (double x);

floor of 2.3 is 2.0

floor of 3.8 is 3.0

floor of -2.3 is -3.0

floor of -3.8 is -4.0

double fmod (double numer , double denom);

fmod of 5.3 / 2 is 1.300000

fmod of 18.5 / 4.2 is 1.700000



Numerical Data Types and Operations

Name	Synonymy	Range	Storage Size
short	short int	-2^{15} (-32,768) to $2^{15}-1$ (32,767)	16-bit signed
unsigned short	unsigned short int	0 to $2^{16}-1$ (65535)	16-bit unsigned
int		-2^{31} (-2147483648) to $2^{31}-1$ (2147483647)	32-bit signed
unsigned int	unsigned	0 to $2^{32}-1$ (4294967295)	32-bit unsigned
long	long int	-2^{31} (-2147483648) to $2^{31}-1$ (2147483647)	32-bit signed
unsigned long	unsigned long int	0 to $2^{32}-1$ (4294967295)	32-bit unsigned
float		Negative range: -3.4028235E+38 to -1.4E-45 Positive range: 1.4E-45 to 3.4028235E+38	32-bit IEEE 754
double		Negative range: -1.7976931348623157E+308 to -4.9E-324 Positive range: 4.9E-324 to 1.7976931348623157E+308	64-bit IEEE 754
long double		Negative range: -1.18E+4932 to 3.37E-4932 Positive range: 3.37E-4932 to 1.18E+4932 Significant decimal digits: 19	80-bit